

“Latest Robots, Especially In Pick-And-Place Applications, Use Machine Vision To Recognise Part Shapes, Colours, And Positions”

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Is it time to invest in robots for a manufacturing setup? Will the returns justify the outlay? Who should evaluate such a decision? To explore these questions, Akanksha Sondhi Gaur from EFY spoke to Epson's Siva Kumar. Here is what he had to say...

Should robotics be viewed primarily as a productivity tool or as a driver of long-term competitiveness?

High-precision manufacturing has become non-negotiable, and manual processes increasingly limit efficiency, consistency, and scale. Automation is therefore not just a means to improve output but a strategic requirement for competitiveness—particularly in India, where initiatives such as Make in India are accelerating manufacturing growth.

The main constraint is skills, as many manufacturers lack hands-on automation experience. We address this gap by guiding customers to identify repetitive tasks and convert them into automated workflows using robots and supporting technologies. This builds internal capability and positions robotics as a sustainable competitive advantage rather than simply a short-term productivity gain.

How does today's robotics differ from 1990s-2000s automation?

Earlier, automation was limited to technician-driven, high-speed tasks in controlled factory settings. Today, robotics has expanded into MSMEs and multiple industries,

with safety as a core design focus. Technologies such as SafeSense define human-safe zones, automatically slowing or stopping robots when people enter the area.

Applications are also more complex, demanding advanced motion control to manage inertia and vibration during high-speed moves, ensuring precise and stable stopping. Modern robotics tightly integrates hardware and software, with platforms such as RC+ enabling easy programming even for non-experts. Peripheral automation, including feeder units, has become essential, while continuous customer feedback shapes solutions—from quick fixes to new technologies developed at the global headquarters level.

Which Indian manufacturing sectors are ready for robotics, and which face hurdles?

We see the communication industry moving fastest in India, with companies actively seeking to automate manual operations. We work closely with these organisations to convert repetitive tasks into automated solutions.

Electric vehicle manufacturers are also growing rapidly, with

applications such as battery welding that require tailored automation. In these industries, many manual processes still exist, so we collaborate with system integrators and manufacturers to address these challenges. Electronics manufacturing is relatively new in India, and we support critical application development from the outset. Across these fast-growing sectors, our approach remains collaborative, combining expertise to enable effective robotics adoption.

Are manufacturers prioritising precision over payload, and how does it shape robot design?

In our market segment, manufacturers prioritise precision, speed, and compact footprint over high payload. We focus on robots under 20kg for tasks requiring high accuracy, such as dynamic-path glue dispensing, where advanced motion control is essential. Our robots integrate sensor technologies, Zero Plus motion control, and SafeSense to deliver stable, vibration-free operation in compact designs. With RC+ software, they are easy to program and can be simulated in 3D, allowing customers to validate performance and flexibility before deployment.